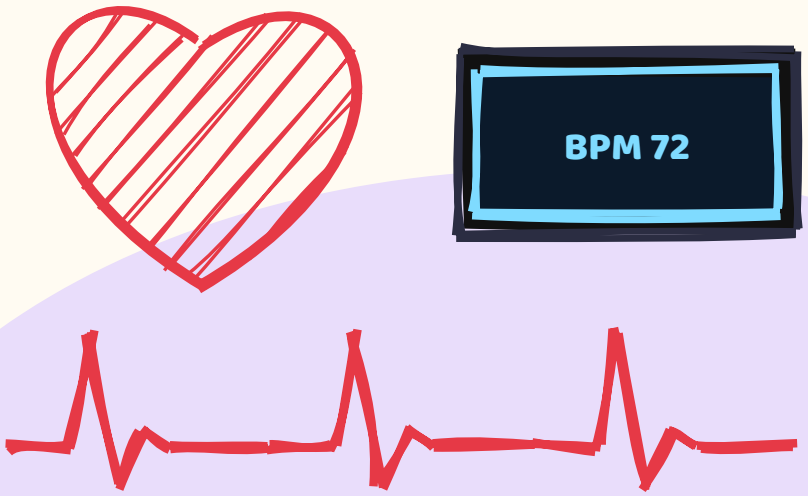
Young Innovators · Class 2

Heart Rate Monitoring System

Measure your heartbeat with light!

Optical pulse sensing **BPM maths** **Live OLED graph**



Did you know?
A smartwatch measures your pulse by shining a green light into your skin, the same trick we use today.

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Let's think...

Can you count your own heartbeat?

Try it now

Put two fingers on your wrist or the side of your neck. Count the beats for **15 seconds**, then multiply by **4**.
Did everyone get the same number? Was it easy to count?



My number was 80!

I lost count...

Let a machine count!

Today: we build a monitor that detects every beat, calculates **beats per minute** and draws a live graph.

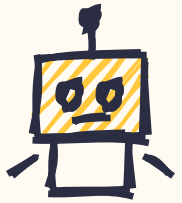
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Learning aims

Today I will learn to...

1. Explain how the heart moves blood and how a sensor can **see** it
2. Read the pulse signal and **detect each beat** with a threshold
3. Calculate **BPM** from the time between beats using millis()
4. Show BPM and a **live graph** on the OLED
5. Discuss **health data privacy** and limits of our device

Keywords
pulse · BPM · signal · threshold · millis() · average · privacy



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Watch and think

Video time!



How Heart Rate Sensors Work in Fitness Bands

Doctor Yogendra · about 5 minutes · youtu.be/QTniLL0sQtA

Watch for...

1. What colour of **light** does the sensor shine into the skin?
2. What changes with every **heartbeat**?
3. Why do readings go wrong when you move a lot?

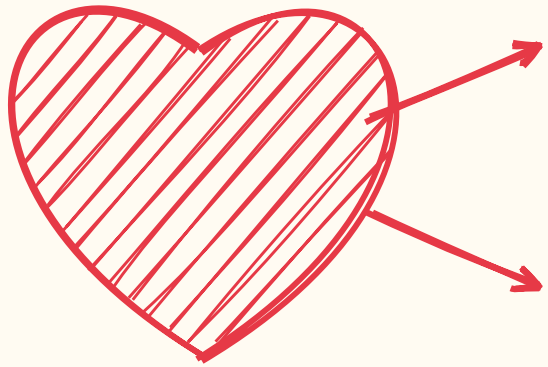
Scan to watch

After the video: Find your pulse on your wrist. Count for 15 seconds and multiply by 4. Compare with the sensor later!

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The science

Every beat is a push of blood



Your heart is a pump

About the size of your fist. Each **squeeze** pushes blood out through your arteries. That push travels all the way to your fingertips.

Who	BPM
Resting teenager	about 60 to 100
After running	can go above 150
A baby	faster, about 100 to 160
Fit athlete at rest	can be under 60

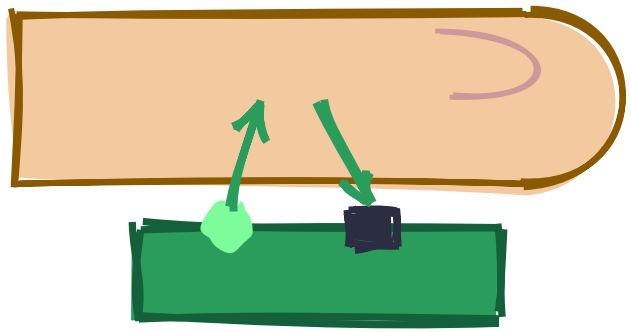
Fingertips swell (a tiny bit!)

With each beat, the tiny blood vessels in your finger fill up a little more, then relax. We can detect that change with **light**.

BPM
Beats Per Minute: how many times the heart beats in 60 seconds.

How the sensor sees

Shine light, measure what bounces back



Step by step

1. A bright **green LED** shines into your skin.
2. Blood **soaks up** green light.
3. A **light sensor** measures what bounces back.
4. More blood (a beat) = **less** light back. The signal makes a wave!

Big word

Photoplethysmography (PPG): measuring blood flow with light. Smartwatches use it!



Real world

Who measures heart rate?



Smartwatches
Track your heart all day and during sport



Hospitals
A finger clip shows pulse and oxygen on a monitor



Athletes
Train in the right heart rate zone



Ambulances
Paramedics check pulse first in an emergency

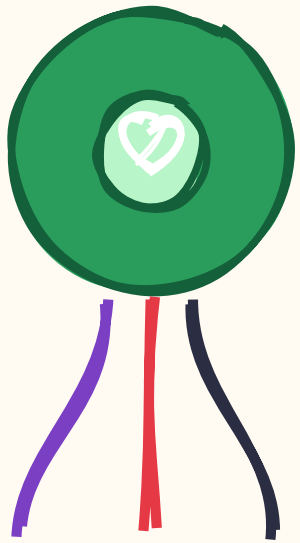
? Has a doctor ever put a clip on your finger? What did the screen show?

Same idea
The hospital finger clip uses red and infrared light instead of green.

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Meet the sensor

The optical pulse sensor



Wire	Uno
S (signal)	A0
+	5V
-	GND

Getting a good signal

Rest your fingertip **lightly**. Pressing hard squashes the blood vessels!
Keep **still**. Moving makes fake beats.
Cover it from bright room light.

Try the earlobe

If fingers are cold, clip it to your **earlobe**. Warm skin gives a stronger signal.

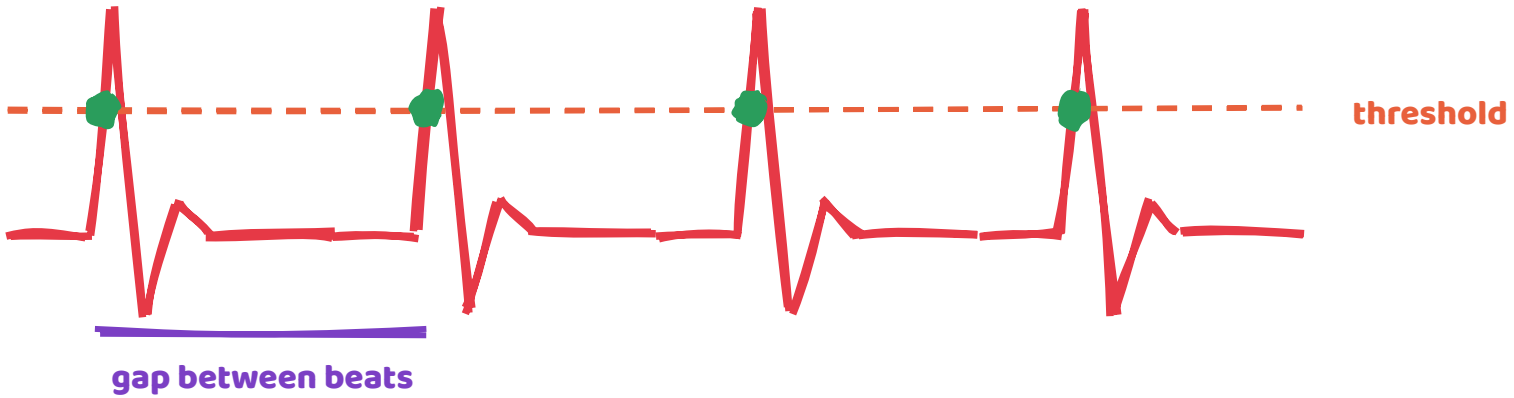
Hygiene
Wipe the sensor between users. Cover the front with clear tape to protect it.

Reading range
analogRead gives 0 to 1023. Resting value is often around 500, beats push it higher.

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Big idea

A beat = crossing the threshold going UP



Beat detection rule

If the signal goes **above** the threshold and we were **not** already in a beat: count a beat! Wait until it drops below again before counting the next one.

Why the 'not already' part?

The wave stays high for several readings. Without it, one beat would be counted 5 times!

Dicrotic notch

The small second bump is the heart valve closing. Our threshold must be ABOVE it.

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Maths moment

From milliseconds to BPM

BPM = 60000 ÷ gap

60000 ms = 1 minute · gap = milliseconds between two beats

millis()

The Uno counts milliseconds since it switched on. Save millis() at each beat. **gap = now - lastBeat**.

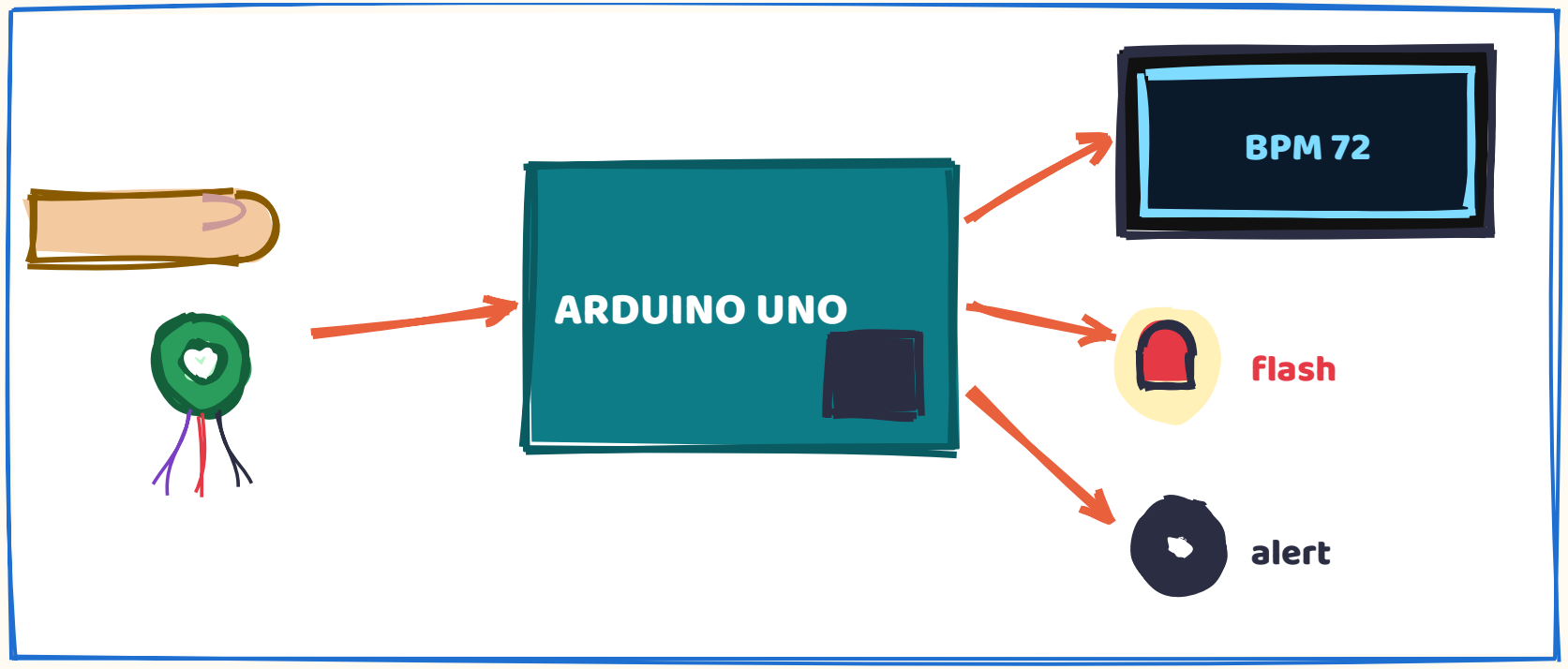
Gap	BPM
1000 ms	60000 ÷ 1000 = **60**
800 ms	60000 ÷ 800 = **75**
500 ms	60000 ÷ 500 = **?**
1200 ms	60000 ÷ 1200 = **?**

Answers

120 BPM and 50 BPM.

The plan

Sense, detect, calculate, show



Outputs

LED flashes with every beat.
OLED shows BPM and a moving graph.
Buzzer beeps if BPM is very high or low.

Serial Plotter

Tools > Serial Plotter in Arduino IDE shows your heartbeat as a live wave!

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Responsible tech

Health data is private data

Your body, your data: heart rate can reveal stress, fitness and health problems.

Ask first: only measure someone who agrees, and let them see their result.

Don't share: never post someone's health numbers without permission.

Big companies: fitness apps collect lots of health data. Who should be allowed to see it?

Not a doctor: our monitor is a school project, NOT a medical device.



Feeling unwell?

If your heart ever feels strange, tell a trusted adult or doctor. Don't rely on a DIY gadget.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Our heart monitor kit (per group)
2. Wiring the heart monitor
3. Detect each beat
4. BPM and a live graph on the OLED
5. Alerts and the Serial Plotter
6. Resting vs active heart rate
7. Heart monitor challenges

Build it

Our heart monitor kit (per group)

Arduino Uno
The brain

Pulse sensor
Sees the beat

OLED 0.96 in
BPM + graph

LED + 220 Ω
Beat flash

Buzzer
Alerts

Arduino IDE 2
Text coding

Also: breadboard, jumper wires, USB cable, laptop, alcohol wipes, a stopwatch for the experiment.

Libraries
Install Adafruit SSD1306 and Adafruit GFX from the Library Manager before class.

MAKE

Build it

Wiring the heart monitor

Part	Uno
Pulse sensor S	A0
Pulse sensor + / -	5V / GND
OLED VCC / GND	5V / GND
OLED SDA / SCL	A4 / A5
LED (+ 220 Ω)	pin 13
Buzzer +	pin 8

Keep the sensor cable still

The sensor wires carry a tiny signal. Tape the cable to the table so it does not wiggle while you measure.

Pin 13

also has the Uno's built-in LED, so you will see a flash even before you add yours.

MAKE

Code it: part 1

Detect each beat

```
const int PULSE = A0, LED = 13;
int threshold = 550;        // tune this!
bool inBeat = false;
unsigned long lastBeat = 0;
int bpm = 0;

void loop() {
  int signal = analogRead(PULSE);
  if (signal > threshold && !inBeat) {
    inBeat = true;
    unsigned long now = millis();
    unsigned long gap = now - lastBeat;
    lastBeat = now;
    if (gap > 300 && gap < 2000) bpm = 60000 / gap;
    digitalWrite(LED, HIGH);
  }
  if (signal < threshold - 20) {
    inBeat = false;
    digitalWrite(LED, LOW);
  }
  delay(10);
}
```

Why 300 and 2000?

A gap under 300 ms would be over 200 BPM, and over 2000 ms is under 30 BPM. Those are almost certainly **noise**, not real beats, so we ignore them.

Why threshold - 20?

The signal wobbles near the line. A small gap (called **hysteresis**) stops one wobble counting as two beats.

MAKE

Code it: part 2

BPM and a live graph on the OLED

```
#include <Adafruit_SSD1306.h>
Adafruit_SSD1306 oled(128, 64, &Wire, -1);
int x = 0;

// inside loop(), after reading signal:
int y = map(signal, 300, 800, 63, 20);
oled.drawPixel(x, constrain(y, 20, 63), WHITE);
x++;
if (x > 127) {            // reached the edge
  x = 0;
  oled.clearDisplay();
}
oled.setCursor(0, 0);
oled.print("BPM ");
oled.print(bpm);
oled.display();
```

How the graph works

Each reading becomes one **dot**. x moves one pixel right each time. map() turns the signal into a height on the screen.

Make it smoother

Show the **average** of the last 5 BPM values. One odd beat will not make the number jump.

setup()
Remember oled.begin(SSD1306_SWITCHCAPVCC, 0x3C) and setTextColor(WHITE).

MAKE

Code it: part 3

Alerts and the Serial Plotter

```
const int BUZZ = 8;
// every 2 seconds:
if (bpm > 0 && (bpm < 50 || bpm > 120)) {
  tone(BUZZ, 2000, 200);    // alert beep
}
Serial.println(signal);     // for Serial Plotter
```

Serial Plotter

Print **only** the signal number, one per line. Then open Tools > Serial Plotter. Find your best threshold by looking at the wave!

Choosing alert limits

We picked 50 and 120 for a **resting** person. During exercise 120 is normal, so a real device needs to know what you are doing!

Tune it

Peaks reach 650? Set the threshold to about 580: above the small bump, below the peaks.

MAKE

Experiment

Resting vs active heart rate

Name	Resting BPM	After 1 min jumping	After 2 min rest
Student A	____	____	____
Student B	____	____	____
Student C	____	____	____
Group average	____	____	____

Predict first

Which number will be biggest? How long until it returns to resting?

Check your machine

Count by hand for 15 s × 4 at the same time. How close is the monitor? A difference of a few BPM is fine.

Safety

Anyone who should not exercise can be the timer or recorder instead.

Level up!

Heart monitor challenges

Choose your challenge

Auto threshold: track the highest and lowest values and set the threshold halfway between.

Heartbeat sound: a short click on the buzzer for every beat.

Recovery timer: after exercise, time how long until BPM drops below 100.

Big heart: draw a heart icon on the OLED that grows with each beat.

Think like an engineer

Why might our monitor give a wrong number for someone with very cold hands, or who is moving?

Boss level

Fit athletes recover faster. Compare recovery times across your class!

Show what you know

What did we learn?

- [x] Each heartbeat pushes blood that a light sensor can see
- [x] A beat = signal crossing the threshold going up
- [x] BPM = 60000 ÷ gap in milliseconds
- [x] Averaging and hysteresis make readings steadier
- [x] Health data is private, and DIY monitors are not medical devices

Exit ticket: if the gap between beats is 750 ms, what is the BPM?

At home

Measure a family member's resting pulse by hand (with permission!). Compare with yours.



Next class:
Smart Classroom Energy Saver!